CBSE MOCK — Pre-Board Practice Series

Central Board of Secondary Education

CLASS XII — PHYSICS

Date: 12 November 2026 | Maximum Marks: 70 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. This question paper contains 5 sections. Read the instructions for each section carefully.
2. All questions are compulsory unless stated otherwise.
3. Write all answers neatly and show relevant working wherever required.
4. Marks for each question are indicated against it.
5. This paper is a Practice Paper — it is similar to but NOT an actual CBSE board paper.
6. Download more FREE practice papers daily at: **www.cbsemock.org.in**

SECTION A — MCQ (16 Marks)

MCQ (1 mark each) | A/R (1 mark each)

Q1. **[1 mark]**

The rms value of an alternating current is 4 A. Its peak value is:

(A) 4 A
(B) 4√2 A
(C) 2√2 A
(D) 8 A

Q2. **[1 mark]**

A conducting wire connects two charged metallic spheres A and B of radii r_1 and r_2. The distance between the spheres is very large compared to their radii. The ratio of electric fields (E_A/E_B) at the surfaces of spheres A and B will be:

(A) r_1/r_2
(B) r_2/r_1
(C) r_1^2/r_2^2
(D) r_2^2/r_1^2

Q. 15 & 16 are Assertion-Reason type. Choose: (A) Both A and R true; R explains A. (B) Both true; R does not explain A. (C) A true, R false. (D) A false, R true.

Q3. **[1 mark]**

Assertion (A): Sky appears blue during daytime. Reason (R): Blue light is scattered least by air molecules.

Q4. **[1 mark]**

The critical angle for a medium with respect to air is θ_c. The refractive index of the medium is:

(A) sin θ_c
(B) cos θ_c
(C) tan θ_c
(D) 1/sin θ_c

Q5. **[1 mark]**

In a transformer, the number of turns in the primary is 500 and in the secondary is 100. If the input voltage is 220 V, the output voltage is:

(A) 1100 V
(B) 44 V
(C) 440 V
(D) 110 V

Q6. **[1 mark]**

The de Broglie wavelength of an electron accelerated through a potential of V volts is proportional to:

(A) V
(B) $\sqrt{V}$
(C) $1/\sqrt{V}$
(D) $1/V$

Q7. **[1 mark]**

The work function of photoelectric effect for a metal is 3.3 eV. The threshold frequency is: ($h = 6.6 \times 10^{-34}$ Js)

(A) 5×10^{14} Hz
(B) 8×10^{14} Hz
(C) 10^{15} Hz
(D) 5×10^{15} Hz

Q8. **[1 mark]**

A nucleus $^{238}_{92}U$ emits an α-particle. The resulting nucleus is:

(A) $^{234}_{90}Th$
(B) $^{234}_{92}U$
(C) $^{238}_{90}Th$
(D) $^{234}_{91}Pa$

Q9. **[1 mark]**

A long straight wire of circular cross-section (radius a) carries a steady current I uniformly distributed across the cross-section. The magnitude of the magnetic field at a point at distance a/2 from the axis will be:

(A) Zero
(B) $\mu_0 I/2\pi a$
(C) $\mu_0 I/4\pi a$
(D) $\mu_0 I/6\pi a$

Q10. **[1 mark]**

In a region, the electric potential varies as $V = 10 - 50x$, where V is in volts and x in meters. The electric field in the region is:

(A) 10 N/C along +x
(B) 10 N/C along −x
(C) 50 N/C along +x
(D) 50 N/C along −x

Q11. **[1 mark]**

In an LCR series circuit, at resonance:

(A) impedance is maximum
(B) current is minimum
(C) power factor is 1
(D) capacitive reactance equals inductance

Q12. **[1 mark]**

Assertion (A): In photoelectric effect, increasing the frequency of incident radiation beyond the threshold frequency increases the kinetic energy of emitted electrons. Reason (R): The energy of a photon is directly proportional to its frequency.

Q13. **[1 mark]**

In a Young's double slit experiment, the slit separation is doubled and the distance from the slits to the screen is halved. The fringe width will:

(A) remain unchanged
(B) double
(C) become half
(D) become one-fourth

Q14. **[1 mark]**

A ray of light is incident at the glass-water interface at an angle i (glass refractive index = 1.5, water = 1.33). The ray will:

(A) always refract into water
(B) undergo total internal reflection for certain values of i
(C) always reflect back into glass
(D) partially reflect and partially refract for all values of i

Q15. **[1 mark]**

According to Bohr's theory, the radius of the nth orbit of hydrogen atom is proportional to:

(A) n
(B) n^2
(C) $1/n$
(D) $1/n^2$

Q16. **[1 mark]**

In a p-n junction diode, the depletion layer is formed due to:

(A) diffusion of majority carriers
(B) drift of minority carriers
(C) drift of majority carriers
(D) diffusion of minority carriers

SECTION B — Very Short Answer (10 Marks)

VSA (2 marks each)

Q17. **[2 marks]**

What is meant by 'bandwidth' of an amplitude modulated wave? If the carrier frequency is 1000 kHz and the modulating signal frequency is 5 kHz, find the bandwidth.

Q18. **[2 marks]**

Define electric flux. Write its SI unit. How is it related to the electric field?

Q19. **[2 marks]**

Draw the V-I characteristics of a p-n junction diode in forward bias and reverse bias. OR What is the difference between intrinsic and extrinsic semiconductors?

Q20. **[2 marks]**

A proton is moving with a velocity $v = 10^5$ m/s and enters a uniform magnetic field $B = 0.2$ T perpendicular to its velocity. Find the radius of the circular path of the proton. (Mass of proton = 1.67×10^{-27} kg, $q = 1.6 \times 10^{-19}$ C) OR State Faraday's laws of electromagnetic induction. Write the expression for induced EMF.

Q21. **[2 marks]**

Draw a labelled diagram of a cyclotron. State the working principle and write the formula for maximum kinetic energy of the accelerated particle.

SECTION C — Short Answer (21 Marks)

SA (3 marks each)

Q22. **[3 marks]**

In hydrogen atom, an electron jumps from 3rd orbit to 1st orbit. Find: (a) The energy of the emitted photon ($E_1 = -13.6$ eV) (b) The wavelength of the emitted radiation

Q23. **[3 marks]**

(a) What is displacement current? Why did Maxwell introduce it? (b) Write the modified Ampere's law including displacement current. OR Derive the relationship between the peak values of electric and magnetic fields in an electromagnetic wave. What is the speed of electromagnetic waves in vacuum?

Q24. **[3 marks]**

Using Gauss's theorem, derive the expression for the electric field intensity at a point due to an infinite plane sheet of charge. What happens to the electric field if the sheet has finite dimensions?

Q25. **[3 marks]**

Explain the phenomenon of polarisation of light. How does Brewster's law relate the polarising angle with the refractive index of the medium? If the polarising angle is 56°, find the refractive index.

Q26. **[3 marks]**

(a) State Ampere's circuital law. (b) Using Ampere's law, derive the magnetic field inside a long solenoid of n turns per unit length carrying current I.

Q27. **[3 marks]**

Explain with the help of a circuit diagram how a full wave rectifier works. Draw the input and output waveforms. OR Explain the working of a transistor as an amplifier (CE configuration) with a circuit diagram.

Q28. **[3 marks]**

A ray of light is incident on a glass-air interface at an angle of 45°. If the refractive index of glass is $\sqrt{2}$, find: (a) The angle of refraction (b) Whether total internal reflection occurs

SECTION D — Case Study (8 Marks)

CBQ (4 marks each — Case Study)

Q29. **[4 marks]**

In 1887, Hertz discovered that when ultraviolet light was shone on a metal surface, electrons were emitted. Einstein explained this using quantum theory in 1905, earning the Nobel Prize in 1921. The photoelectric equation is: KE_max = hν − φ, where φ is the work function and h is Planck's constant (6.63×10^{-34} Js). (a) State Einstein's photoelectric equation and explain each term. (b) The stopping potential for a metal with work function 2 eV is 1 V. Find the frequency of incident radiation. (c) Why does increasing the intensity of light not affect the kinetic energy of photoelectrons? OR (c) The photoelectric threshold wavelength of silver is 3250 Å. Find its work function.

Q30. **[4 marks]**

Semiconductors are materials with electrical conductivity between conductors and insulators. Pure silicon has a conductivity of about 10^{-3} S/m. When impurities are added (doping), the conductivity increases dramatically. In n-type semiconductors, donor impurities (like phosphorus) are added, while in p-type semiconductors, acceptor impurities (like boron) are added. The energy gap in silicon is 1.1 eV. (a) What is meant by 'energy gap' in a semiconductor? (b) Distinguish between n-type and p-type semiconductors. (c) Why does the conductivity of a semiconductor increase with temperature? OR (c) Explain how a depletion region is formed in a p-n junction.

SECTION E — Long Answer (15 Marks)

LA (5 marks each)

Q31. **[5 marks]**

(a) Derive the expression for the energy stored in a capacitor of capacitance C charged to voltage V. (b) Three capacitors of 2μF, 3μF, and 6μF are connected: (i) in series — find equivalent capacitance (ii) in parallel — find equivalent capacitance (c) A parallel plate capacitor with air between the plates has a capacitance of 8 pF. What will be the capacitance if the distance between the plates is reduced by half and the space between plates is filled with a medium of dielectric constant 6? OR (a) State Kirchhoff's laws. Apply them to find the current in each branch of the given circuit (Wheatstone bridge arrangement). (b) Calculate the equivalent resistance of the network if three resistors of 6Ω each are connected in a triangle and a 3Ω resistor connects the two vertices.

Q32. **[5 marks]**

(a) Explain the principle of a moving coil galvanometer. Derive the expression for the deflection produced. (b) How is a galvanometer converted to: (i) an ammeter? Write the formula for the shunt resistance. (ii) a voltmeter? Write the formula for the series resistance. (c) A galvanometer has resistance 20Ω and shows full-scale deflection for 1 mA. How will you convert it to an ammeter to measure 5 A? OR (a) State the laws of electromagnetic induction. (b) Derive an expression for the self-inductance of a solenoid of n turns per unit length, length l, and cross-sectional area A. (c) A circular coil of 100 turns and radius 10 cm is placed in a uniform magnetic field of 0.5 T. If the coil is rotated with an angular velocity of 100 π rad/s, find the maximum induced EMF.

Q33. **[5 marks]**

(a) State Huygens' principle. Use it to derive Snell's law of refraction. (b) In a Young's double slit experiment, the slit separation is 0.5 mm and the screen is placed 1 m away. If the wavelength of light used is 600 nm, find: (i) The fringe width (ii) The position of 3rd bright fringe from centre OR (a) What is diffraction of light? Give two differences between interference and diffraction. (b) A single slit of width 0.1 mm is illuminated by light of wavelength 600 nm. The diffraction pattern is observed on a screen 1 m away. Find the width of the central maximum.

✦ END OF QUESTION PAPER ✦

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